Range Sum Query 2D (Immutable)

→ Given a 2D matrix, for an input of top-left & bottom-right corners, I have to return the sum enclosed by the area in O(1) time

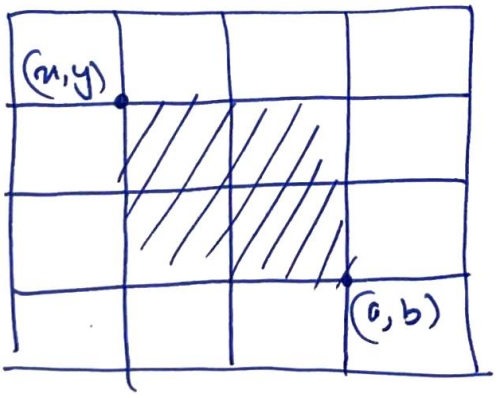
→ To ensure O(1) time, it suggests that some pre-computation be done, such that our effective query can be made efficiently.

My approach (prefix-sum)

→ I construct a row-wise prefix sum in a new matrix from the original matrix (because the question mentions immutable, ∴ I did not mutate the input matrix)

→ Now, for the given top-left & bottom right corners, fetch the sum of each enclosed row, within the columnar limits, total it, & return sum

(M×N)



$S_1 = A[x][b] - A[x][y-1]$

↳ $y \geq 0$

$S_1 = A[x][b]$

↳ $y = 0$

↳ calculate & add S_1 of each enclosing row